

F954 — The curved sea is built, on the real D_{IV}^5 operator. Keeper was right: the operator was ours all along — Paper 118's Bergman-Dirac, and written as the Dolbeault operator $\bar{\partial} + \bar{\partial}^\dagger$ it sidesteps the spin connection entirely. I built the sea on it, and idempotency + Krein hold by construction. ★ The operator is Paper 118's Bergman-Dirac = the Dolbeault (Kähler-Dirac) operator $D = \bar{\partial} + \bar{\partial}^\dagger$ on D_{IV}^5 . Dolbeault spinor $S = \Lambda^\bullet T^{\{0,1^*\}}$, $\dim 2^{\{n_C\}} = 32$; Clifford $\gamma^{\{z_i\}} = \sqrt{2} \cdot \text{wedge}(dz^i)$, $\gamma^{\{\bar{z}_j\}} = \sqrt{2} \cdot \text{contract}$; chirality $\Gamma_5 = (-1)^{\{\text{degree}\}}$, $16 \oplus 16$. No spin connection needed — $\bar{\partial}$ is canonical (the connection is packaged in $K^{\{1/2\}}$). I rebuilt the 32×32 Clifford and verified all 75 anticommutation relations to 1.3×10^{-14} (Paper 118 T2365, reproduced). ★ The sea = $\chi_{(-\infty, 0)}(D)$, and all the credential properties hold by construction: (1) idempotent — the negative-spectral projector satisfies $\Lambda_-^2 = \Lambda_-$ to 3×10^{-15} , spectrally, no gluing. (2) half-filled — rank $16/32$, the physical Dirac sea (exactly the chirality split). (3) Krein by construction — $\Gamma_5 D \Gamma_5 = -D$ exactly (the Bergman-Dirac anticommutes with its own chirality), so Γ_5 is the grading and the sea/anti-sea are Krein-conjugate; the Krein-symmetry I proved to 15 places in F951 is now *structural* on the real operator, not just the flat model. ★ Well-posedness — the gap is real and it's ours. The Wallach K-type spectrum $\lambda^2(m_1, m_2) = m_1(m_1 + n_C) + m_2(m_2 + N_C) - n_C \cdot g/4$ has exactly three negative- D^2 K-types — $(0,0)$, $(0,1)$, $(1,0)$ at $\{-8.75, -4.75, -2.75\}$ — created by the Lichnerowicz shift $n_C \cdot g/4 = 35/4 = 8.75$ (the BST primary product in the geometry itself). Three is odd $\rightarrow \nu(M) = 1$, the Möbius $Z/2$ class. This shift is the Kähler+quaternionic gap around zero that makes χ_- well-posed, exactly as Keeper said. ★ What this closes and what remains. *Closed*: the sea is built on the genuine D_{IV}^5 operator (not the flat Minkowski model of F952), idempotent + Krein by construction, Clifford + gap verified. The construction is end-to-end: Paper 118's operator $\rightarrow \chi_- \rightarrow$ the fermionic projector, all three credential properties by spectral+chirality structure. *Remaining*: the causal two-point kernel $P(x,y)$ on the domain (the mode functions away from the origin, Paper 118 Section 9.1, multi-week) — that is where the Lorentzian spacelike/timelike census lands, and F952's flat-Minkowski demonstration already showed it comes out right (spacelike $L=0$, timelike real). So: the algebraic/Krein half is on the real operator now; the Lorentzian-causal half is proven in the flat model; the synthesis (curved + Lorentzian two-point kernel) is the last integral. ★ The weight, in the Dolbeault picture (progress, still the open fork). The fermion bundle is $S = \Lambda^\bullet \otimes K^{\{1/2\}}$: the $K^{\{1/2\}}$ gives the half-form weight $\text{genus}/2 = n_C/\text{rank} = 5/2$ (grounded, F950), and the Dolbeault degree adds integer steps. The physical fermion's degree fixes the shift — degree-1 gives $5/2 + 1 = 7/2$, degree-0 gives $5/2$. This is the reproducing fork in geometric clothing: which Dolbeault degree carries the physical fermion. I

The sea on the real D_{IV}^5 operator (verified)

property	value	mechanism
Clifford (75 relations)	1.3×10^{-14}	Paper 118 T2365, Dolbeault wedge/contract
idempotent $\Lambda_-^2 = \Lambda_-$	3×10^{-15}	spectral projector of D (by construction)
half-filled	16/32	physical Dirac sea = chirality split
Krein: $\Gamma_5 D \Gamma_5 = -D$	0 (exact)	Bergman-Dirac anticommutes with its chirality
gap: negative- D^2 K-types	exactly 3, Lichnerowicz shift 35/4	Kähler+quaternionic well-posedness

No gluing, no spin connection, on the genuine operator. The F951 sandwich failure is fully bypassed: a spectral projector is idempotent for free.

What's proven vs what's the last integral

- Proven (real operator): χ_- (Bergman-Dirac) is idempotent + Krein; the gap is well-posed.
- Proven (flat model, F952): the sea's Lorentzian causal structure — spacelike $L=0$, timelike real.
- The last integral: the domain two-point kernel $P(x,y)$ (mode functions, Paper 118 Sec 9.1) — unites the two halves into the curved-Lorentzian object; then Elie's three tests run on the genuine domain object and B1's credential is claimable on D_{IV}^5 .

The weight — Dolbeault degree is the fork

$S = \Lambda^\bullet \otimes K^{\{1/2\}}$; weight = genus/2 (= 5/2, grounded) + Dolbeault degree. The fork “6 vs 7” (full exponent) = “5/2 vs 7/2” (half) = which Dolbeault degree carries the physical fermion. Not closed; but now a degree count on the real operator, blind to data (Grace/Cal, joint).

RUBRICS Layer-2 done-bar (curved sea, honest)

- ☑ Identified Paper 118's Bergman-Dirac as the Dolbeault operator $\partial + \partial^\dagger$ (spin-connection-free); rebuilt the 32×32 Clifford, 75 relations to $1e-14$.
- ☑ Built the sea = $\chi_-(D)$: idempotent $3e-15$ (spectral), half-filled 16/32, Krein $\Gamma_5 D \Gamma_5 = -D$ exact — all by construction, on the real operator.
- ☑ Verified the well-posedness gap: exactly 3 negative- D^2 Wallach K-types, Lichnerowicz shift $n_C \cdot g/4 = 35/4$.
- ☑ Held the tier: algebraic/Krein half on the real operator; Lorentzian-causal half in the flat model (F952); the domain two-point kernel is the remaining integral (Paper 118 Sec 9.1). B1 credential not yet claimed on D_{IV}^5 .
- ☑ Recast the weight fork as a Dolbeault-degree count on the real operator (blind); not closed. Nothing pushed.

Handoffs

- @Elie — the sea is on the real D_{IV}^5 operator now (`dolbeault_sea(pc)` in the reference impl): idempotent $3e-15$, half-filled 16/32, Krein exact. Your three tests need the domain two-point kernel (the mode functions, Paper 118 Sec 9.1) — that's the last integral before the Lorentzian census runs on the genuine object. Until then: the flat sea (F952) is your validated positive control (all three pass), and this Dolbeault build is the curved operator confirmed idempotent+Krein. Charm/top weight-7 check still a good lull task.
- @Keeper — built, on Paper 118, as you called it. The Bergman-Dirac written as $\partial + \partial^\dagger$ sidesteps the spin connection; the sea $\chi_-(D)$ is idempotent ($3e-15$) + Krein ($\Gamma_5 D \Gamma_5 = -D$ exact) by construction, on the real operator; the Lichnerowicz gap ($n_C \cdot g/4 = 35/4$, exactly 3 negative- D^2 K-types) is the well-posedness you flagged.

Remaining: the domain two-point kernel (Paper 118 Sec 9.1, the mode functions) for the Lorentzian causal census — the last integral, F952 shows it comes out right. Also: Paper 118's correction note asked me to re-derive the KE scalar-curvature / D-normalization against genus=5 — the scalar curvature $R = -n_C \cdot g = -35$ and the Lichnerowicz $n_C \cdot g/4$ are consistent (they use $n_C \cdot g$, not the kernel exponent); the only genus-5 fix is the *kernel* exponent line (already corrected). I'll confirm in the v0.3 pass.

- @Grace / @Cal — the weight is now a Dolbeault-degree count on the real operator. $S = \Lambda^\bullet \otimes K^{\{1/2\}}$: half-form weight = genus/2 = 5/2 (grounded), and the physical fermion's Dolbeault degree sets the shift (degree 1 $\rightarrow$ 7/2, degree 0 $\rightarrow$ 5/2; the “6 vs 7” full-exponent fork). This is more tractable than the abstract discrete-series parameter — it's counting which degree carries the fermion, blind to the up-quark. Genus = 5 stands. Three rejections still held.
- @Cal — the curved-sea cold-read (your §448 chain): χ_- (Bergman-Dirac) idempotent 3e-15, Krein $\Gamma_5 D\Gamma_5 = -D$ exact, on the real 32-dim Dolbeault operator (75 Clifford relations 1e-14). Hostile question for you: is the algebraic (origin/flat-momentum) construction a fair stand-in for the curved domain two-point kernel, or can the Bergman metric variation (Paper 118 Sec 9.1) break idempotency or the gap? And the weight: reject 7 until the Dolbeault degree is counted — not $g=7$, not data, not $5+2$.
- @Casey — the sea is built on our own operator, and it holds. The thing that turns our object into Finster's — the filled negative-energy sea — I built it this time not on a flat stand-in but on *our* operator, the one that was sitting in Paper 118 the whole time. Keeper's insight saved weeks: instead of the hard “spinor derivative” everyone dreads, our operator can be written with the *natural* calculus of a complex space (the ∂ operator), which needs no extra machinery. On that operator the sea is a genuine projector (do-it-twice-get-the-same, to fifteen places) and it has the space-and-time symmetry built in (our object anticommutes with its own mirror, exactly). And the “gap” that makes the whole thing well-defined turned out to be one of our own numbers — 35/4, which is (five times seven) over four, sitting right there in the geometry. What's left is one honest integral — writing the object as a function of *two* points across the whole space, not just at the center — and that's the piece that lets Elie run the final space-vs-time test on the real domain. The flat version already passed it. And the seven: I turned it into a cleaner question — counting a “degree” on our own operator — but I did not answer it, and I won't say I did. Nothing pushed.

Notes only; no toy/theorem claimed (curved-sea construction on Paper 118 operator). F954: CURVED SEA built on Paper 118 Bergman-Dirac = Dolbeault operator $\partial + \bar{\partial}^\dagger$ (spin-connection-FREE, Kähler-Dirac). Dolbeault spinor $\Lambda^{(C)5} = 32$, $\gamma^\wedge\{z\} = \sqrt{2}$ wedge, $\gamma^\wedge\{zbar\} = \sqrt{2}$ contract, chirality $\Gamma_5 = (-1)^{\deg} 16 \oplus 16$. Rebuilt 32×32 Clifford, 75 relations 1.3e-14 (T2365). SEA = $\chi_-(D)$: (1) IDEMPOTENT $\Lambda_-^2 = \Lambda_-$ 3e-15 (spectral, no gluing); (2) HALF-FILLED 16/32 (physical Dirac sea); (3) KREIN $\Gamma_5 D\Gamma_5 = -D$ EXACT (0.0) $\rightarrow$ chirality grading, F951 Krein-symmetry now structural on real operator. GAP: exactly 3 negative- D^2 Wallach K-types (0,0),(0,1),(1,0) = {−8.75, −4.75, −2.75}, Lichnerowicz shift $n_C \cdot g/4 = 35/4 = 8.75 \rightarrow$ Kähler+quaternionic well-posedness (Keeper). CLOSED: sea on GENUINE D_{IV}^5 operator, idempotent+Krein by construction. REMAINING: domain two-point kernel $P(x,y)$ (mode functions, Paper 118 Sec 9.1, multi-week) = Lorentzian causal census; F952 flat showed it comes out right (spacelike $L=0$ /timelike real). WEIGHT: $S = \Lambda^{*\otimes K}\{1/2\}$, weight=genus/2 (5/2 grounded) + Dolbeault DEGREE; fork 6vs7 = which degree carries physical fermion (degree1 $\rightarrow$ 7/2, degree0 $\rightarrow$ 5/2); NOT closed but now degree-count on real operator blind. Paper118 correction: KE scalar $R = -n_C \cdot g = -35$ + Lichnerowicz $n_C \cdot g/4$ consistent (use $n_C \cdot g$ not kernel exponent); only kernel-exponent line needed genus-5 fix (done). Ref impl: `dolbeault_clifford()`, `dolbeault_sea()`. Elie: sea on real operator, needs 2-pt kernel for tests, flat=positive control. Keeper: built on 118 as called; gap=35/4; remaining=2pt integral. Grace/Cal: weight=Dolbeault degree count blind. Cal: cold-read origin-vs-curved fairness; reject 7 until degree counted. Nothing pushed. — Lyra